1) (a) (8 points) Approximate the integral

$$\int_0^2 (3x - x^3)\, dx$$

using the Trapezoid Rule with $n = 4$. Express your answer as a fraction.

$\Delta x = \frac{2-0}{4} = \frac{1}{2}$

$T_4 = \frac{\Delta x}{2}\left[f(0) + 2f(\tfrac{1}{2}) + 2f(1) + 2f(\tfrac{3}{2}) + f(2)\right]$

$= \frac{1}{4}\left[0 + 2(\tfrac{11}{8}) + 2(2) + 2(\tfrac{9}{8}) - 2\right]$

$= \frac{1}{4}\left[0 + \tfrac{11}{4} + 4 + \tfrac{9}{4} - 2\right]$

$= \frac{1}{4}\left[5 + 4 - 2\right]$

$= \boxed{\frac{7}{4}}$

(b) (12 points) Evaluate the following improper integrals, or show they diverge:

(i) $\int_3^{19} \frac{1}{\sqrt[4]{x-3}}\, dx = \lim_{t \to 3^+} \int_t^{19} (x-3)^{-1/4}\, dx$

$= \lim_{t \to 3^+} \frac{4}{3}(x-3)^{3/4} \Big|_t^{19}$

$= \frac{4}{3} \lim_{t \to 3^+} \left[16^{3/4} - (t-3)^{3/4}\right]$

$= \frac{4}{3}\left[8 - 0\right] = \boxed{\frac{32}{3}}$

(ii) $\int_1^{\infty} \frac{1}{(3x+1)^{3/5}}\, dx = \lim_{t \to \infty} \int_1^t \frac{1}{(3x+1)^{3/5}}\, dx$

$u = 3x + 1$

$du = 3dx$

$= \frac{1}{3} \lim_{t \to \infty} \int_4^{3t+1} u^{-3/5}\, du$

$= \frac{1}{3} \lim_{t \to \infty} \frac{5}{2} u^{2/5} \Big|_4^{3t+1}$

$= \frac{5}{6} \lim_{t \to \infty} \left[(3t+1)^{2/5} - 4^{2/5}\right]$

$= \frac{5}{6}\left[\infty - 4^{2/5}\right]$

$\boxed{\text{diverges}}$

2) (a) (10 points) Find the length of the curve

$$y = \frac{1}{3}(x^2+2)^{3/2}, \quad 0 \le x \le 3$$

$$y' = \frac{1}{2}(x^2+2)^{1/2}(2x) = x\sqrt{x^2+2}$$

$$\sqrt{1+(y')^2} = \sqrt{1+x^2(x^2+2)} = \sqrt{x^4+2x^2+1} = \sqrt{(x^2+1)^2} = |x^2+1|$$
$$= x^2+1$$

$$L = \int_0^3 (x^2+1)\,dx$$
$$= \frac{1}{3}x^3 + x \Big|_0^3$$
$$= 9+3-(0+0)$$
$$= \boxed{12}$$

(b) (10 points) Find the surface area of the surface obtained by revolving the curve

$$y = \sqrt{6x+1}, \quad 1 \le x \le 9$$

about the x-axis.

radius $= y = \sqrt{6x+1}$

arc length thing $= \sqrt{1+(y')^2} = \sqrt{1+\left[\frac{1}{2}(6x+1)^{-1/2}(6)\right]^2} = \sqrt{1+\left(\frac{3}{\sqrt{6x+1}}\right)^2}$

$$= \sqrt{1+\frac{9}{6x+1}}$$

$$= \sqrt{\frac{6x+1+9}{6x+1}}$$

$$= \sqrt{\frac{6x+10}{6x+1}}$$

$$A = 2\pi\int_1^9 \sqrt{6x+1}\sqrt{\frac{6x+10}{6x+1}}\,dx = 2\pi\int_1^9 \sqrt{6x+10}\,dx$$

$u = 6x+10$

$du = 6dx$

$$= \frac{\pi}{3}\int_{16}^{64} \sqrt{u}\,du$$

$$= \frac{\pi}{3}\cdot\frac{2}{3}u^{3/2}\Big|_{16}^{64} = \frac{2\pi}{9}(512-64) = \frac{2\pi}{9}(448) = \boxed{\frac{896\pi}{9}}$$

3) (a) (12 points) Find the limits of the following sequences, or show that they diverge.

(i) $\{\ln(4n^3+1) - \ln(2n^2+5) - \ln n\}$ $= \ln\left(\frac{4n^3+1}{2n^3+5n}\right)$

$\frac{4n^3+1}{2n^3+5n} = \frac{4+\frac{1}{n^3}}{2+\frac{5}{n^2}} \to \frac{4}{2} = 2$, so $\ln\left(\frac{4n^3+1}{2n^3+5n}\right) \to \boxed{\ln 2}$

OR:

$\frac{4n^3+1}{2n^3+5n} \approx \frac{4n^3}{2n^3} = 2$ when n is large

(ii) $\left\{\frac{(-1)^n n^2}{n^2+1}\right\}$

$\left|\frac{(-1)^n n^2}{n^2+1}\right| = \frac{n^2}{n^2+1} = \frac{1}{1+\frac{1}{n^2}} \to \frac{1}{1} = 1 \neq 0$, so $\frac{(-1)^n n^2}{n^2+1}$ alternates btwn. terms that $\to 1$ and terms that $\to -1$.

$\boxed{\text{diverges}}$

(iii) $\left\{\frac{n^4+2}{\sqrt{9n^8+n^6+4n^2-1}}\right\}$

$= \frac{1+\frac{2}{n^4}}{\sqrt{9+\frac{1}{n^2}+\frac{4}{n^6}-\frac{1}{n^8}}} \to \frac{1}{\sqrt{9}} = \boxed{\frac{1}{3}}$

OR:

$\approx \frac{n^4}{\sqrt{9n^8}} = \frac{n^4}{3n^4} = \frac{1}{3}$ when n is large

(b) (8 points) The following series converges. Find its sum.

$\sum_{n=1}^{\infty} \frac{4^n+(-1)^n}{5^n} = \sum_{n=1}^{\infty}\left(\frac{4}{5}\right)^n + \sum_{n=1}^{\infty}\left(-\frac{1}{5}\right)^n$

$= \left(\frac{4}{5}+\frac{16}{25}+\frac{64}{125}+\dots\right) + \left(-\frac{1}{5}+\frac{1}{25}-\frac{1}{125}+\dots\right)$

$a=\frac{4}{5}, r=\frac{4}{5}$ $\qquad$ $a=-\frac{1}{5}, r=-\frac{1}{5}$

$= \frac{4}{5}\cdot\frac{1}{1-\frac{4}{5}} - \frac{1}{5}\cdot\frac{1}{1+\frac{1}{5}}$

$= \frac{4}{5}(5) - \frac{1}{5}\left(\frac{5}{6}\right)$

$= 4 - \frac{1}{6}$

$= \boxed{\frac{23}{6}}$

4) (30 points) Determine whether the following series converge or diverge.

(a) $\sum_{n=1}^{\infty} \frac{n}{\sqrt{n^5+1}}$

Compare to $\sum \frac{n}{n^{5/2}} = \sum \frac{1}{n^{3/2}}$, which converges.

$\sqrt{n^5+1} > \sqrt{n^5}$

$\frac{1}{\sqrt{n^5+1}} < \frac{1}{\sqrt{n^5}}$

$\frac{n}{\sqrt{n^5+1}} < \frac{n}{\sqrt{n^5}} = \frac{1}{n^{3/2}}$, so $\boxed{\text{converges}}$ by Comparison Test

(Limit Comparison Test would also work)

(b) $\sum_{n=2}^{\infty} \frac{1}{n\,(\ln n)^4}$

Integral Test:

$$\int_2^{\infty} \frac{1}{x(\ln x)^4}\,dx = \lim_{t\to\infty} \int_2^t \frac{1}{x(\ln x)^4}\,dx$$

$u = \ln x$

$du = \frac{1}{x}\,dx$

$$= \lim_{t\to\infty} \int_{\ln 2}^{\ln t} u^{-4}\,du$$

$$= \lim_{t\to\infty} -\frac{1}{3}u^{-3}\Big|_{\ln 2}^{\ln t}$$

$$= -\frac{1}{3}\lim_{t\to\infty}\left[\frac{1}{(\ln t)^3} - \frac{1}{(\ln 2)^3}\right]$$

$$= -\frac{1}{3}\left[0 - \frac{1}{(\ln 2)^3}\right]$$

converges, so the series $\boxed{\text{converges}}$ by the Integral Test

(c) $\sum_{n=1}^{\infty} \left(\frac{2n^3}{n^3+4}\right)^n$

Root Test: $\sqrt[n]{|a_n|} = \frac{2n^3}{n^3+4} \to 2 > 1$

$\boxed{\text{diverges}}$ by Root Test

(d) $\sum_{n=0}^{\infty} \frac{n^2+1}{\sqrt{4n^4+n^2+3}}$

$$\frac{n^2+1}{\sqrt{4n^4+n^2+3}} = \frac{1+\frac{1}{n^2}}{\sqrt{4+\frac{1}{n^2}+\frac{3}{n^4}}} \to \frac{1}{\sqrt{4}} = \frac{1}{2} \neq 0$$

OR: $\approx \frac{n^2}{\sqrt{4n^4}} = \frac{n^2}{2n^2} = \frac{1}{2}$ when n is large

$\boxed{\text{diverges}}$ by Test for Divergence

(e) $\sum_{n=1}^{\infty} \frac{n!}{3\cdot 8\cdot 13\cdots(5n-2)}$

Ratio Test:

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(n+1)!}{3\cdot 8\cdot 13\cdots(5n-2)(5n+3)} \cdot \frac{3\cdot 8\cdot 13\cdots(5n-2)}{n!}\right|$$

$$= \frac{n+1}{5n+3}$$

$$\to \frac{1}{5} < 1$$

$\boxed{\text{converges}}$ by Ratio Test

5) **(10 points) Determine whether the following series are absolutely convergent, conditionally convergent, or divergent.**

(a) $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}\, n^2}{4^n}$

Root Test: (or can use Ratio Test)

$$\sqrt[n]{|a_n|} = \left(\frac{n^2}{4^n}\right)^{1/n} = \frac{n^{2/n}}{4} \to \frac{1}{4} < 1$$

$\boxed{\text{absolutely convergent}}$

(b) $\sum_{n=2}^{\infty} \frac{(-1)^n\, n^2}{\sqrt{n^5-3}} \approx \sum \frac{(-1)^n}{n^{1/2}}$ when n is large

$\sum |a_n| = \sum \frac{n^2}{\sqrt{n^5-3}}$ diverges by comparison with $\sum \frac{1}{n^{1/2}}$

but $\sum a_n = \sum \frac{(-1)^n n^2}{\sqrt{n^5-3}}$ converges by Alt. Series Test

$\boxed{\text{conditionally convergent}}$

6) (10 points) The alternating series

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,4^n}$$

converges. Estimate the sum to within 0.001. (Write your answer as a fraction.)

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,4^n} = 1 - \frac{1}{1(4)} + \frac{1}{2(16)} - \frac{1}{6(64)} + \underbrace{\frac{1}{24(256)}}_{< \frac{1}{1000}} - \cdots$$

so

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{n!\,4^n} \approx 1 - \frac{1}{4} + \frac{1}{32} - \frac{1}{384}$$

$$= \frac{384 - 96 + 12 - 1}{384}$$

$$= \boxed{\frac{299}{384}}, \text{ correct to within } \frac{1}{24(256)} < 0.001$$